5) Creat structure variable student and estimate size of structure manually. Then write a C program to find size of structure using 'sizeof' program. Now observe both the outputs. Are they same or different? Justify your answer.

A) Structure Definition:

```
Struct student
{
    int rollno;
    char grade;
    float marks;
};
```

Size estimation

int roll = 4 bytes
char grade = 1 byte
float marks = 4 bytes
Size = 4+1+4 = 9 bytes.

C Program using sizeof

```
#include<stdio.h>
struct student
{
    int rollno;
    char grade;
    float marks;
};
int main()
{ struct student s;
  Printf("Size = %.2u bytes\n",sizeof(s));
    return 0;
}
```

o/p: size = 12 bytes.

∴ The o/p are different because compiler adds padding btw structure members so that data is stored at memory addresses that are efficient for processor to access. Although actual data occupies 9 bytes, compiler inserts 3 extra bytes, making it 12 bytes.